

Build AI-Ready Search: Integrating Docling with OpenSearch for Advanced RAG and Agentic Applications



Phil Nash, IBM



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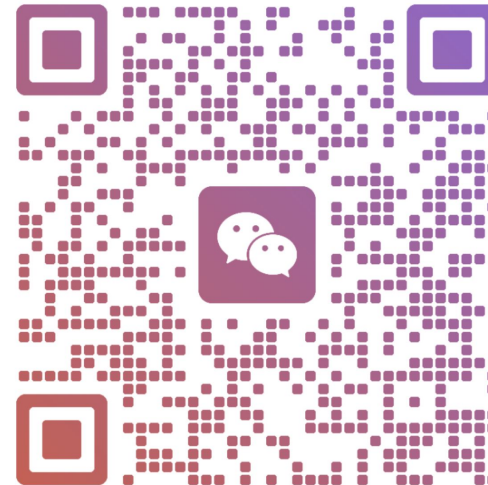


Shanghai, China



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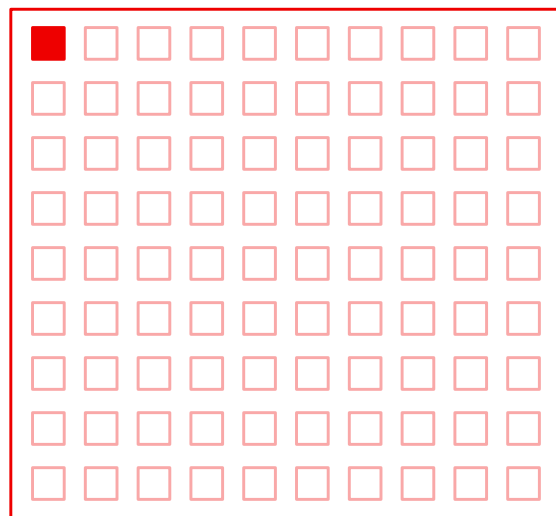
Phil Nash

 Developer relations engineer at IBM

 <https://philna.sh>



Knowledge/data cutoff for LLMs



Less than 1% of all enterprise data is represented in foundation models

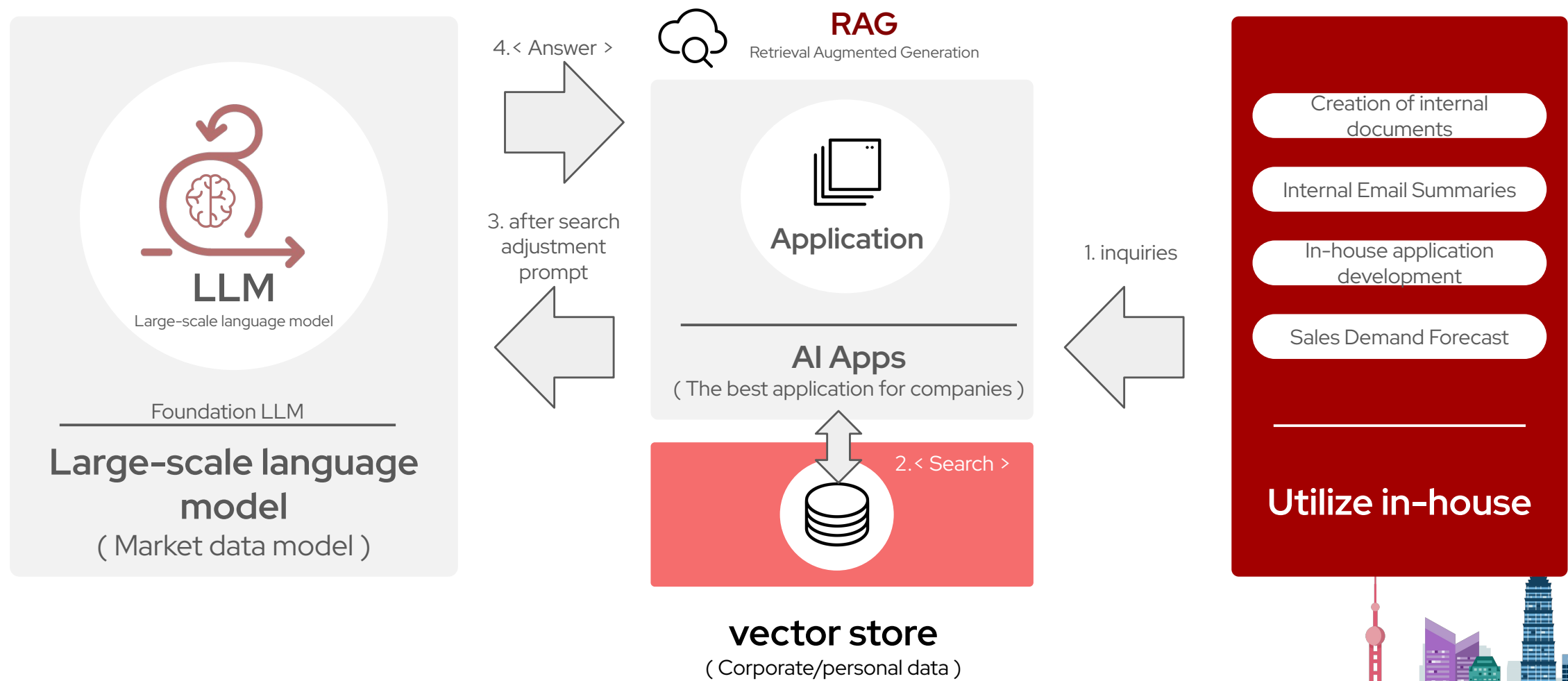
Enterprise organizations need to

1. Start from a trusted base model
2. Create a new representation of their data(*)
3. Deploy, scale, and create value with their AI

* Much of the data is stuck in PDFs



RAG (Retrieval-Augmented Generation)



OpenSearch



“Vegetative electron microscopy”

 **gurovdigital** 15 h ...
lol, over 20 scientific papers now feature the nonsensical term ‘vegetative electron microscopy’.

all because an AI misinterpreted a 1959 article, merging ‘vegetative’ and ‘electron microscopy’ from separate columns.

enzyme present in spores. that the composition of *B.* and from spore coats of *B.* a composition similar to the cell wall, from the results of Norris of Leeds University (1959). He treated spores with preparation of lytic enzymes and examined the electron microscopy. No exospore was obtained. It was not known whether in spores, or another enzyme for lysis of the sporangial cells. When thick suspensions of *B. cereus* were



Date syrup (as one of the agricultural wastes) was used to produce bacterial cellulose using *Gluconastobacter xylinus*. Fourier transform infrared spectroscopy (FTIR), vegetative electron microscopy, and X-ray diffraction were used to determine the structure of bacterial cellulose, cellulose fibers, and crystallinity of the samples (Moosavi and Yousefi, 2011). After 14 days of incubation at 28 °C, the highest yield of cellulose

Silver and gold nanoparticles for antimicrobial purposes against multi-drug resistance bacteria [HTML] m
N Rabiee, S Ahmadi, O Akhavan, R Luque - Materials, 2022 - mdpi.com
... Dead bacteria have been observed by imaging and elemental analysis using transmission electron microscopes (TEM), vegetative electron microscopy, and EDX (X-Ray Probe Microscopy) ...
☆ Cited by 112 Related articles

“vegetative electron microsc...”
Scholar YEAR
Study of CNT@ Fe3O4 effects on *Aeromonas hydrophila* and *Yersinia ruckeri* bacteria isolated from fish.
M Alishiri, KR Tavabe, A Mirvaghefi... - Journal of ..., 2019 - search.ebscohost.com
... -carbon nanotubes synthesized by spectroscopic and microscopic techniques including X-ray diffraction spectrum, shaking sample magnetometer and vegetative electron microscopy ...
☆ Cited by 1 Related articles
Green synthesis of silver nanoparticles via *Ganoderma lucidum* fungus extract and its antibacterial effects on *Klebsiella pneumoniae* isolates from urinary tract ...
M Jamshidian-Mojaver, M Amir... - Alborz University ..., 2021 - ...
... Vegetative electron microscopy was used to measure the ...
☆ Cited by 1 Related articles
METALLOGRAPHIC STUDIES OF IRAN'S IRON AGE: CASE STUDY BRONZE PIECES FROM JEYRÂN TEPE, OZBAKI [PDF] res
B SODAEI, H RAHNEMA - researchgate.net
This study is a report of the results of metallographic study of 5 bronze pieces found in Jeyrân Tepe dating back to the Iron

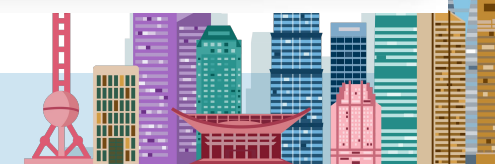
Well, actually... (sneak peek)

were incubated with an extract from spores disintegrated at pH 7.0. Peptide was released which established that the coats contained substrate for the lytic enzyme present in spores. Peptide was also released from spore coats of *B. megaterium* by the action of the enzyme from *B. cereus* spores. The lytic enzyme did not attack intact resting spores.

The spore develops in the vegetative cell, which thus becomes a sporangium. It is by no means certain what happens to the vegetative cell wall when the spore is released. In *Clostridium* species it appears that at least part of this structure is retained as an outer membrane around the spore. It is the opinion of some workers that the wall of the sporulating cell forms the exosporium which exists as an outer

characteristic type. It was concluded that at least part of the sporangial wall was dissolved away to allow release of the spore. It appears likely that the exosporium of *B. cereus* does not have a composition similar to that of the vegetative cell wall, from the results obtained by Dr. J. R.

The spore develops in the vegetative cell, which thus becomes a sporangium. It is by no means certain what happens to the vegetative cell wall when the spore is released. In *Clostridium* species it appears that at least part of this structure is retained as an outer membrane around the spore. It is the opinion of some workers that the wall of the sporulating cell forms the exosporium which exists as an outer coat around spores of several *Bacillus* species. Spores of several varieties of *B. cereus* had exosporia whereas these structures appeared to be absent from spores of *B. megaterium* and *B. subtilis*. It seems, however, that in *Bacillus* species at least, the greater part of the vegetative cell wall is dissolved away before the developed spore is released. If this is true, then soluble components containing the characteristic constituents should appear in the medium during spore release. Culture filtrates from *B. cereus* organisms at various stages of growth and sporulation were hydrolyzed and the hydrolyzates analyzed for amino sugars and diaminopimelic acid (28). Results showed that a large increase in the concentration of these substances in the culture filtrate occurred during spore release (table 2); they were found to be present in a nondialyzable peptide of the characteristic type. It was concluded that at least part of the sporangial wall was dissolved away to allow release of the spore. It appears likely that the exosporium of *B. cereus* does not have a composition similar to that of the vegetative cell wall, from the results obtained by Dr. J. R. Norris of Leeds University (personal communication). He treated spores with a highly active preparation of lytic enzyme from *B. cereus* spores and examined the effect by means of electron microscopy. No evidence of lysis of the exosporium was obtained.



Structured content in PDFs

KDD '22, August 14–18, 2022, Washington, DC, USA Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, and Peter Staar

Table 1: DocLayNet dataset overview. Along with the frequency of each class label, we present the relative occurrence (as % of row "Total") in the train, test and validation sets. The inter-annotator agreement is computed as the mAP@0.5-0.95 metric between pairwise annotations from the triple-annotated pages, from which we obtain accuracy ranges.

class label	Count	% of Total			triple inter-annotator mAP @ 0.5-0.95 (%)									
		Train	Test	Val	All	Fin	Man	Sci	Law	Pat	Ten			
Caption	22524	2.04	1.77	2.32	84-89	40-61	86-92	94-99	95-99	69-78	n/a			
Footnote	6318	0.60	0.31	0.58	83-91	n/a	100	62-88	n/a	n/a	82-97			
Formula	25027	2.25	1.90	2.96	83-85	n/a	84-87	86-96	n/a	n/a	n/a			
List-item	185660	17.19	13.34	15.82	87-88	74-83	90-92	97-97	81-85	75-88	93-95			
Page-footer	70878	6.51	5.58	6.00	93-94	88-90	95-96	100	92-97	100	96-98			
Page-header	58022	5.10	6.70	5.06	85-89	66-76	90-94	98-100	91-92	97-99	81-86			
Picture	45976	4.21	2.78	5.31	69-71	56-59	82-86	69-82	80-95	66-71	59-76			
Section-header	142884	12.60	15.77	12.85	83-84	70-81	90-92	94-95	87-94	69-73	78-86			
Table	34733	3.20	2.27	3.60	77-81	75-80	83-86	98-99	58-80	79-84	70-85			
Text	510377	45.82	49.28	45.00	84-86	81-86	88-93	89-93	87-92	71-79	87-95			
Title	5071	0.47	0.30	0.50	60-72	24-63	30-63	94-100	82-96	68-79	24-56			
Total	1107470	941123	99816	66531	82-83	71-74	79-81	89-94	86-91	71-76	68-85			

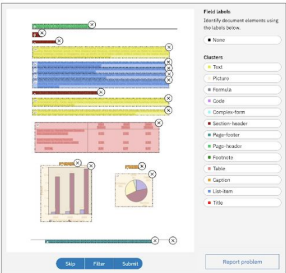


Figure 3: Corpus Conversion Service annotation user interface. The PDF page is shown in the background, with overlaid text-cells (in darker shades). The annotation boxes can be drawn by dragging a rectangle over each segment with the respective label from the palette on the right.

we distributed the annotation workload and performed continuous quality controls. Phase one and two required a small team of experts only. For phases three and four, a group of 40 dedicated annotators were assembled and supervised.

Phase 1: Data selection and preparation. Our inclusion criteria for documents were described in Section 3. A large effort went into ensuring that all documents are free to use. The data sources

include publication repositories such as arXiv⁵, government offices, company websites as well as data directory services for financial reports and patents. Scanned documents were excluded wherever possible because they can be rotated or skewed. This would not allow us to perform annotation with rectangular bounding-boxes and therefore complicate the annotation process.

Preparation work included uploading and parsing the sourced PDF documents in the Corpus Conversion Service (CCS) [22], a cloud-native platform which provides a visual annotation interface and allows for dataset inspection and analysis. The annotation interface of CCS is shown in Figure 3. The desired balance of pages between the different document categories was achieved by selective subsampling of pages with certain desired properties. For example, we made sure to include the title page of each document and bias the remaining page selection to those with figures or tables. The latter was achieved by leveraging pre-trained object detection models from PubLayNet, which helped us estimate how many figures and tables a given page contains.

Phase 2: Label selection and guideline. We reviewed the collected documents and identified the most common structural features they exhibit. This was achieved by identifying recurrent layout elements and lead us to the definition of 11 distinct class labels. These 11 class labels are *Caption*, *Footnote*, *Formula*, *List-item*, *Page-footer*, *Page-header*, *Picture*, *Section-header*, *Table*, *Text*, and *Title*. Critical factors that were considered for the choice of these class labels were (1) the overall occurrence of the label, (2) the specificity of the label, (3) recognisability on a single page (i.e. no need for context from previous or next page) and (4) overall coverage of the page. Specificity ensures that the choice of label is not ambiguous, while coverage ensures that all meaningful items on a page can be annotated. We refrained from class labels that are very specific to a document category, such as *Abstract* in the *Scientific Articles* category. We also avoided class labels that are tightly linked to the semantics of the text. Labels such as *Author* and *Affiliation*, as seen in DocBank, are often only distinguishable by discriminating on

⁵<https://arxiv.org/>

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% of Total

triple inter-annotator mAP @ 0.5-0.95 (%)

[...]
Count
22524
6318
25027
185660
70878
58022
45976
142884
34733
510377
5071
1107470

! Tables not understood

! Image content missing

! Line wraps not understood

! Multi-column often breaks order

! undesired page headers

- ✓ Very fast and cheap
- ✗ Incomplete
- ✗ Loss of structure
- ✗ Noisy
- ➡ Unfit for most use cases



Docling

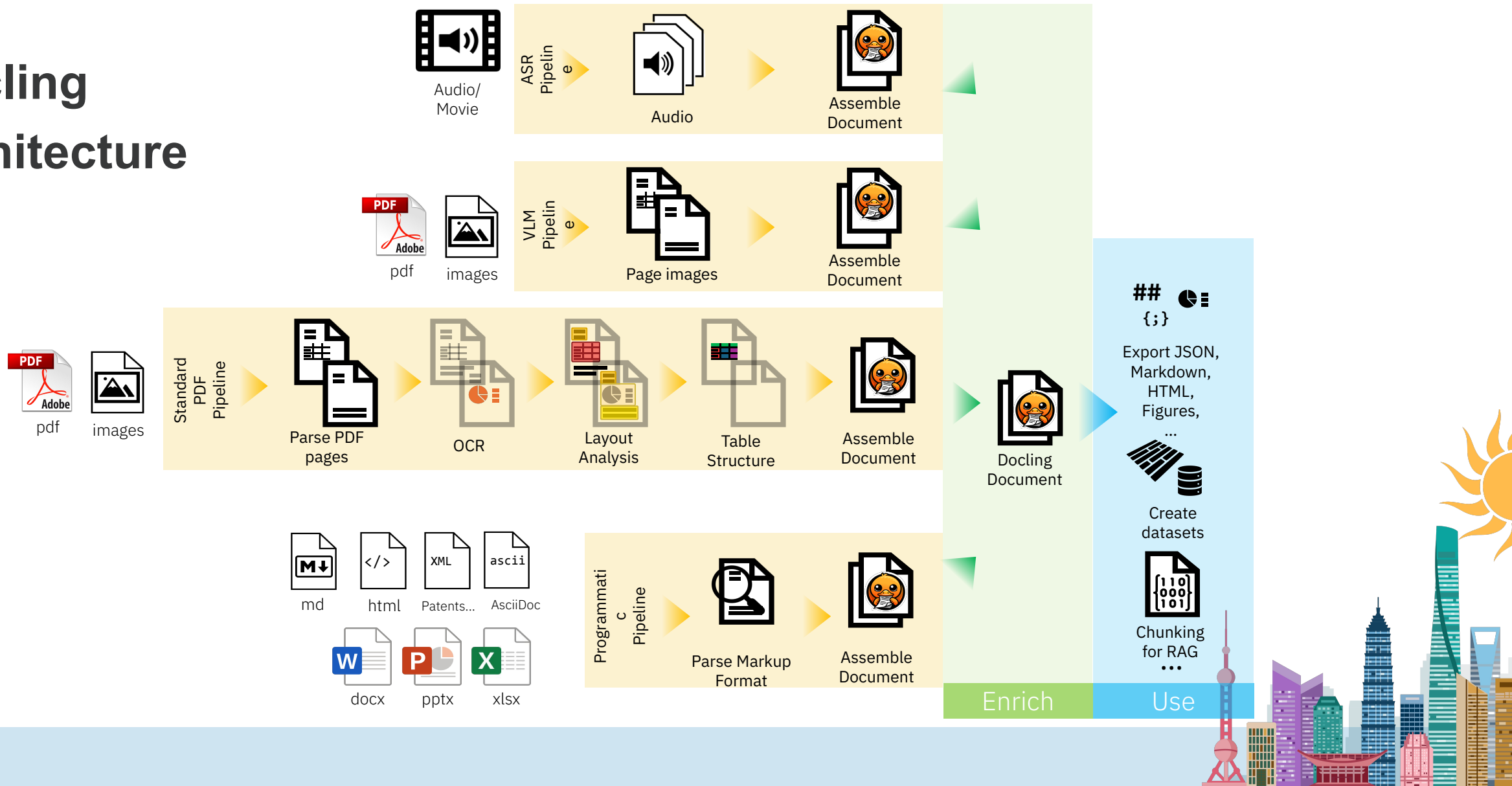
-  Multiple parsing / export formats
-  Advanced PDF understanding
-  Unified DoclingDocument representation format
-  Local execution for sensitive data



LF AI
& DATA



Docling architecture



Docling CLI

```
> pip install docling
```

Installation options: https://docling-project.github.io/docling/getting_started/installation/

```
> docling path/to/horrible.pdf
```

CLI reference: <https://docling-project.github.io/docling/reference/cli/>



Docling in Python

```
from docling.document_converter import DocumentConverter

source = "https://arxiv.org/pdf/2408.09869"
converter = DocumentConverter()
doc = converter.convert(source).document
print(doc.export_to_markdown())
```



Docling UI

```
> pip install docling-serve  
> docling-serve run --enable-ui
```



Docling Hybrid Chunker

Naive Splitting

...Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. The results are shown in the table below.

Column A	
Column A	Column B
...	...
...	...

Breaks context, splits tables, confuses retrieval.

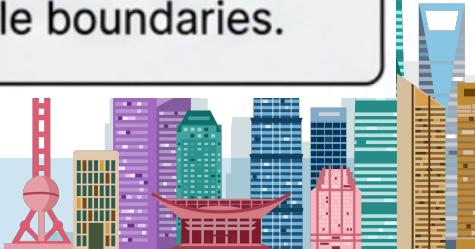
Docling Hybrid Chunking

Header

...Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. The results are shown in the table below.

Column A	Column B
Table	

Respects headers, lists, and table boundaries.



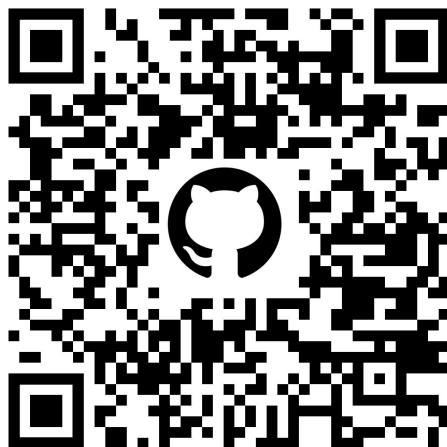
What do we need for AI ready search?

Text extracted (images described, table structure retained)

Turned into chunks

Indexed in OpenSearch





<https://github.com/philnash/opensearch-docling-node>

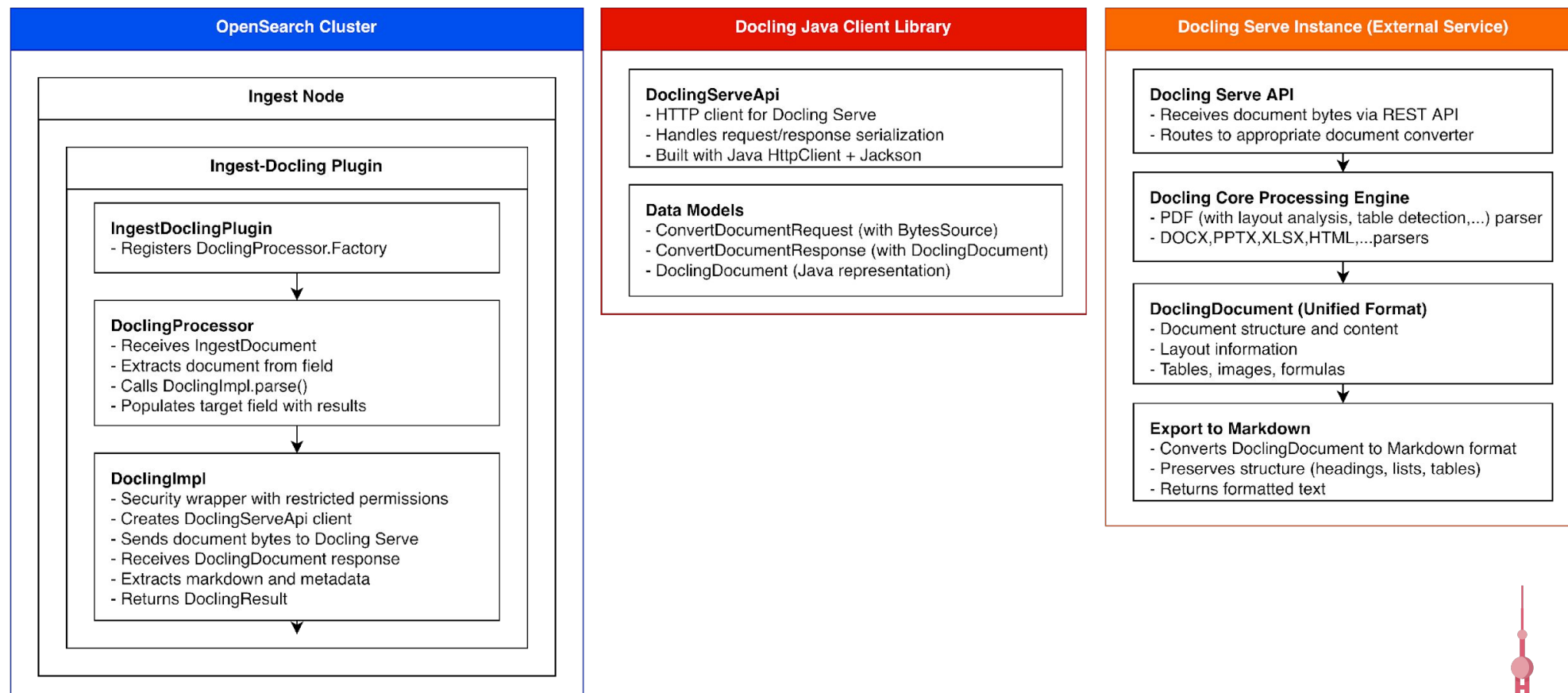


https://github.com/docling-project/docling/blob/main/docs/examples/rag_opensearch.ipynb



Tighter integration with Docling Java API

A custom ingestion plugin leverages Docling Java API to process several document types into a rich, unified format (**DoclingDocument**) that can be used in OpenSearch ingestion pipelines (expected release: 2Q 2026)





OpenRAG



 OpenSearch

 Langflow



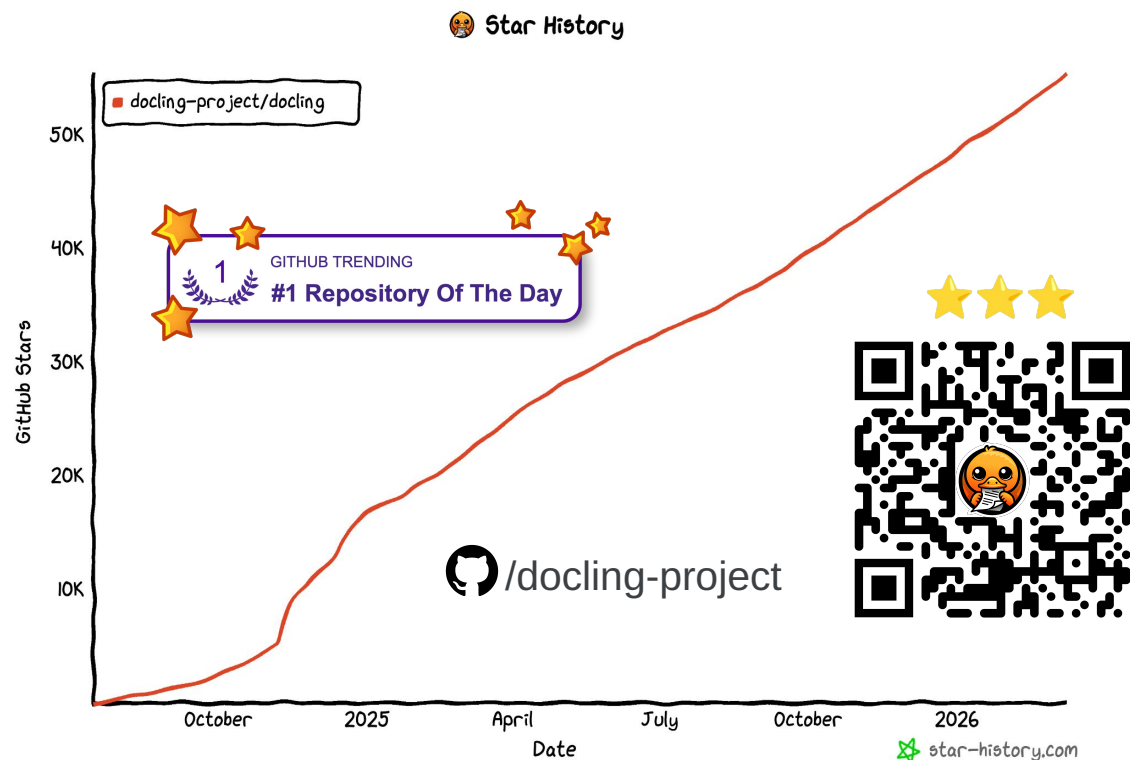
Please ★ and contribute!

<https://github.com/langflow-ai/openrag>



Docling project

- Community adoption
- ★ 55k+ [GitHub](#) stars
- 🐍 4.7M+ downloads last month from [PyPI](#)



docling

Mar 10, 2025 → Mar 10, 2026

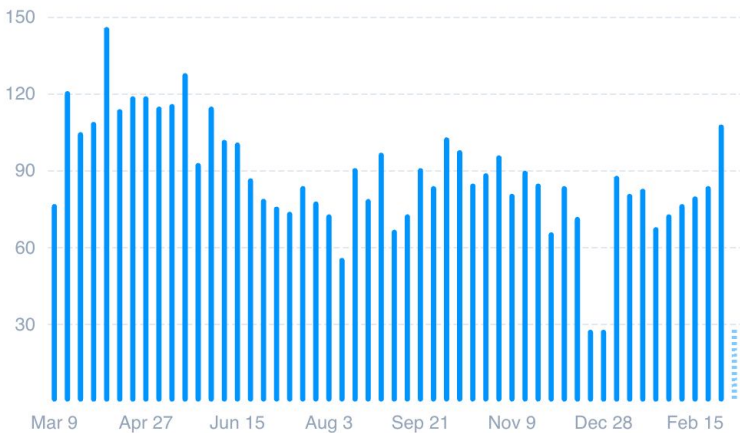
Active contributors

2,186 📈 168.6% (+1,372) vs. 814 last period

Maintainers
15

Reviewers
58

weekly active contributors



OLFX Insights

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- Model runtimes and presets
- Metrics packages
- Memory hardening and profiling
- Enhanced large scale processing
- XBRL input support
- docling-slim package
- Parser improvements

<https://lfai.data.foundation/projects/docling/>

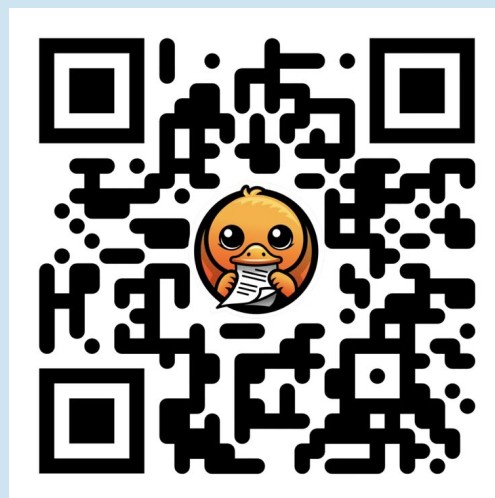




Thank you!

谢谢！

Kiitos!



docling.ai

